

CSSS/SOC/STAT 321: Problem Set #3

Due: February 29, 2024

For this problem set, we will analyze data from the following article:

Fraga, Bernard. (2015) “Candidates or Districts? Reevaluating the Role of Race in Voter Turnout,” *American Journal of Political Science*, Vol. 60, No. 1, pp. 97–122.

Fraga assesses the theory that minority voters are more likely to vote in elections featuring co-ethnic candidates. He shows that the association between minority voter turnout and co-ethnic candidates disappears once we take into account district-level racial composition. In particular, he demonstrates that in districts where blacks make up a greater share of the voting-age population, blacks in that district are more likely to vote in elections *regardless* of candidate race. Although the main analyses he carries out are a bit more complicated than what we cover in *QSS*, we can replicate his substantive findings using the multiple regression approach we’ve learned.

A description of the variables is listed below:

Name	Description
year	Year the election was held
state	State in which the election was held
district	District in which the election was held (unique within state but not across states)
turnout	The proportion of the black voting-age population in a district that votes in the general election
CVAP	The proportion of a district’s voting-age population that is black
candidate	Binary variable coded “1” when the election includes a black candidate; “0” when the election does not include a black candidate

Question 1

Fraga analyzes turnout data for four different racial and ethnic groups, but for this analysis we will focus on the data for black voters. Load `blackturnout.csv`. Which years are included in the dataset? How many different states are included in the dataset?

Loading the dataset

```
rm(list = ls())
library(readr)
blackturnout <- read_csv("blackturnout.csv")

## New names:
## Rows: 1237 Columns: 7
## -- Column specification
## ----- Delimiter: "," chr
## (1): state dbl (6): ...1, year, district, turnout, CVAP, candidate
## i Use 'spec()' to retrieve the full column specification for this data. i
```

```
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * '' -> '...1'
```

Examining the dataset:

```
head(blackturnout)
```

```
## # A tibble: 6 x 7
##   ...1 year state district turnout CVAP candidate
##   <dbl> <dbl> <chr>    <dbl>    <dbl> <dbl>    <dbl>
## 1     1     2008 AK         0    0.710 0.0350      0
## 2     2     2010 AK         0    0.448 0.0323      0
## 3     3     2010 AK         1    0.448 0.0323      0
## 4     4     2008 AK         1    0.710 0.0350      0
## 5     5     2006 AK         1    0.439 0.0318      0
## 6     6     2010 AL         0    0.397 0.256      0
```

Examining the dataset: years

```
summary(blackturnout$year)
```

```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   2006   2006   2008   2008   2010   2010
```

```
unique(blackturnout$year)
```

```
## [1] 2008 2010 2006
```

Examining the dataset: states

```
unique(blackturnout$state)
```

```
## [1] "AK" "AL" "AR" "AZ" "CA" "CO" "CT" "DE" "FL" "GA" "IA" "IL" "IN" "KS" "KY"
## [16] "LA" "MA" "MD" "ME" "MI" "MN" "MO" "MS" "NC" "NE" "NH" "NJ" "NM" "NV" "NY"
## [31] "OH" "OK" "OR" "PA" "RI" "SC" "TN" "TX" "UT" "WA" "WI" "WV"
```

```
length(unique(blackturnout$state))
```

```
## [1] 42
```

Answer 1

The years 2006, 2008, and 2010 are included in the dataset. There are 42 states in the dataset.

Question 2

Create a boxplot that compares turnout in elections with and without a co-ethnic candidate. Be sure to use informative labels. Interpret the resulting graph.

```
library(scales)
```

```
##
```

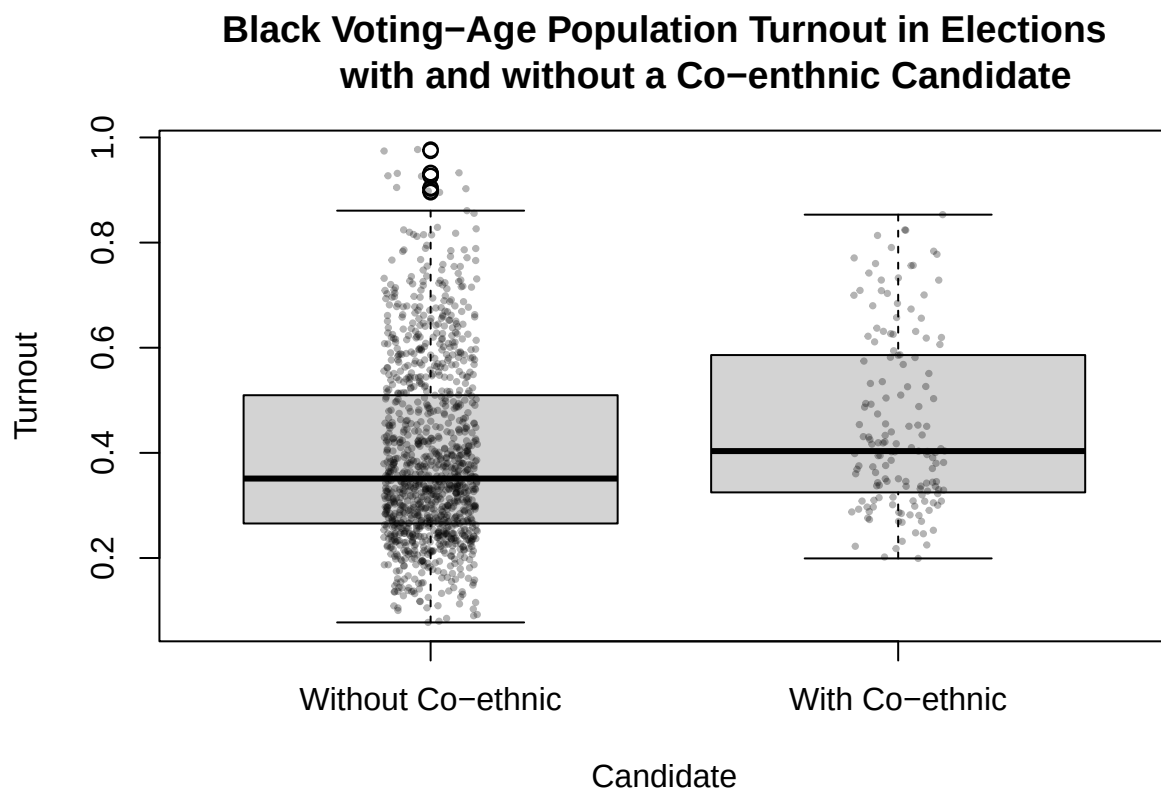
```
## Attaching package: 'scales'
```

```
## The following object is masked from 'package:readr':
```

```
##
```

```
## col_factor
```

```
boxplot(turnout ~ candidate, data = blackturnout,
        xlab = "Candidate",
        ylab = "Turnout",
        main = "Black Voting-Age Population Turnout in Elections
with and without a Co-ethnic Candidate",
        names = c("Without Co-ethnic", "With Co-ethnic")
)
stripchart(turnout ~ candidate, vertical = TRUE, data = blackturnout,
           method = "jitter", add = TRUE, pch = 20, cex = 0.55, col = alpha('black', 0.3))
```



Answer 2

The median turnout of black voting-age population in elections *with* a co-ethnic candidate (a little over 40%) is slightly higher than that in elections *without* a co-ethnic candidate (a little under 40%). There are more elections *without* co-ethnic candidates.

Question 3

Run a linear regression with black turnout as your outcome variable and candidate co-ethnicity as your predictor. Report the coefficient on your predictor and the intercept. Interpret these coefficients. Do not merely comment on the direction of the association (i.e., whether the slope is positive or negative). Explain what the value of the coefficients mean in terms of the units in which each variable is measured. Based on these coefficients, what would you conclude about blacks voter turnout and co-ethnic candidates?

```
lm_cand_turn <- lm(candidate ~ turnout, data = blackturnout)
coef(lm_cand_turn)
```

```
## (Intercept)      turnout
##  0.03165126  0.21930700
```

Answer 3

The y-intercept, 0.032, means that in elections with no co-ethnic candidates, the model predicts that 3.2% of the black voting-age will turn out.

The coefficient on co-ethnic candidates is 0.22, which means that in elections with a co-ethnic candidate, turnout will be around 22 percentage points higher on average.

In elections with a co-ethnic candidate, the predicted black voting-age population turnout is around 25.1%.

Black voter turnout seems to be slightly higher when there is a co-ethnic candidate in the election.

Question 4

You decide to investigate the results of the previous question a bit more carefully because the elections with co-ethnic candidates may differ from the elections without co-ethnic candidates in other ways. Create a scatter plot where the x-axis is the proportion of co-ethnic voting-age population and the y-axis is black voter turnout. Color your points according to candidate co-ethnicity. That is, make the points for elections featuring co-ethnic candidates one color, and make the points for elections featuring no co-ethnic candidates a different color. Interpret the graph.

```
blackturnout$col[blackturnout$candidate == 0] <- "darkcyan"
```

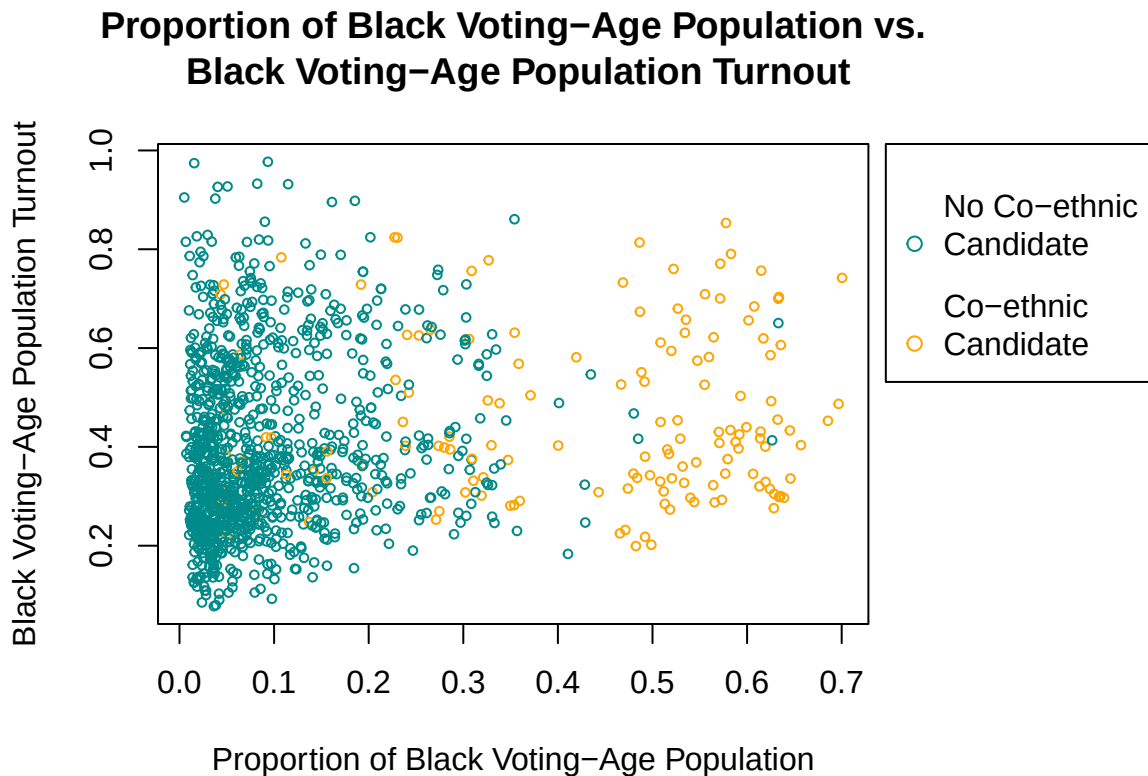
```
## Warning: Unknown or uninitialised column: 'col'.
```

```
blackturnout$col[blackturnout$candidate == 1] <- "orange"
```

```
par(mar=c(5, 5, 5, 9), xpd=TRUE)
```

```
plot(x = blackturnout$CVAP, y = blackturnout$turnout, col = blackturnout$col,
     pch = 1, cex = 0.6, xlab = "Proportion of Black Voting-Age Population",
     ylab = "Black Voting-Age Population Turnout",
     main = "Proportion of Black Voting-Age Population vs. Black Voting-Age Population Turnout")
```

```
legend("topright", inset=c(-0.4, 0), legend=c("No Co-ethnic\nCandidate", "Co-ethnic\nCandidate"), pch=c(1, 1))
```



Answer 4

Based on the scatter plot, co-ethnic candidates tend to run in districts with higher proportions of black voting-age population, and non-co-ethnic candidates tend to run in districts with very low proportions of black voting-age population. There is a slight positive relationship (0.168) between the proportion of black voting-age population and black-voting age population turnout.

Question 5

Run a linear regression with black turnout as your outcome variable and with candidate co-ethnicity and co-ethnic voting-age population as your predictors. Report the coefficients, including the intercept. Interpret the coefficients on the two predictors, ignoring the intercept for now (you will interpret the intercept in the next question). Explain what each coefficient represents in terms of the units of the relevant variables.

```
lm_cdist_turn <- lm(turnout ~ candidate + CVAP, data = blackturnout)
coef(lm_cdist_turn)
```

```
## (Intercept)    candidate      CVAP
## 0.375275442 -0.007364317 0.207391943
```

Answer 5

The coefficient of a co-ethnic candidate is -0.007, meaning that holding black voter population proportion constant, elections with a co-ethnic candidate will have a black voter turnout that is 0.7 percentage points

(a fairly small number) lower on average among black voters.

The coefficient of black voting-age population proportion in a district is 0.207. When the proportion of black voters increases by 0.1 (10 percentage points), black voter turnout is predicted to be 2.1 percentage points higher. In a district with a population of completely black voters with no co-ethnic candidates, the predicted turnout is around 58.2%.

Question 6

Now interpret the intercept from the regression model with two predictors. Is this intercept a substantively important or interesting quantity? Why or why not?

Answer 6

The intercept is 0.375. The model predicts that the black voting-age population turnout for elections with no co-ethnic candidate and districts with no black voting-age population is around 37.5%, which is not a quantity of interest because that is not possible (cannot have a black voting-age turnout of 37.5% if there are no black-voting age people in a district) and not relevant to what we are trying to observe (effects of candidate co-ethnicity and proportion of district population on black voter turnout).

Question 7

Based on the regression model with one predictor, what do you conclude about the relationship between co-ethnic candidates and black voter turnout? Based on the regression model with two predictors, what do you conclude about the relationship between co-ethnic candidates and black voter turnout? Please ignore issues of statistical significance for this question.

Answer 7

Based on the regression model with one predictor, the relationship between co-ethnic candidates and black voter turnout seems to be positive, with turnout being 22 percentage points higher for elections featuring co-ethnic candidates.

Based on the regression model with two predictors, the relationship between co-ethnic candidates is less pronounced and slightly negative, with turnout being 0.7 percentage points lower for elections featuring co-ethnic candidates when the proportion of black voters in a district is held constant. Instead, it seems that the proportion of black voters in a district has a greater effect on black voter turnout than co-ethnicity of the candidate in an election.